

ContractIQ Database Design - Sprint 3

Database Schema

users

Column	Data Type	Key	Description
id	INTEGER	PK	User ID
full_name	VARCHAR(100)		Name
email	VARCHAR(255)	UNIQUE	Email
password	VARCHAR(255)		Password
role	VARCHAR(50)		Role
is_active	BOOLEAN		Status

Purpose: Stores user accounts

contracts

Column	Data Type	Key	Description
id	INTEGER	PK	Contract ID
user_id	INTEGER	FK->users.id	Owner
title	VARCHAR(255)		Title
start_date	DATE		Start
end_date	DATE		End
status	VARCHAR(50)		Status

Purpose: Stores contracts

contract_versions

Column	Data Type	Key	Description
id	INTEGER	PK	ID
contract_id	INTEGER	FK->contracts.id	Contract
version_no	INTEGER		Version
file_path	TEXT		File

Purpose: Stores versions

obligations

Column	Data Type	Key	Description
id	INTEGER	PK	ID
contract_id	INTEGER	FK->contracts.id	Contract
description	TEXT		Description
due_date	DATE		Due
status	VARCHAR(50)		Status

Purpose: Tracks obligations

renewals

Column	Data Type	Key	Description
id	INTEGER	PK	ID

contract_id	INTEGER	FK->contracts.id	Contract
renewal_date	DATE		Renewal
status	VARCHAR(50)		Status

Purpose: Stores renewals

notifications

Column	Data Type	Key	Description
id	INTEGER	PK	ID
user_id	INTEGER	FK->users.id	User
message	TEXT		Message
is_read	BOOLEAN		Read

Purpose: Stores notifications

reports

Column	Data Type	Key	Description
id	INTEGER	PK	ID
user_id	INTEGER	FK->users.id	User
report_type	VARCHAR(100)		Type
generated_at	TIMESTAMP		Generated

Purpose: Stores reports

audit_logs

Column	Data Type	Key	Description
id	INTEGER	PK	ID
user_id	INTEGER	FK->users.id	User
action	VARCHAR(255)		Action
created_at	TIMESTAMP		Time

Purpose: Stores logs

activities

Column	Data Type	Key	Description
id	INTEGER	PK	ID
user_id	INTEGER	FK->users.id	User
contract_id	INTEGER	FK->contracts.id	Contract
activity	VARCHAR(255)		Activity
created_at	TIMESTAMP		Time

Purpose: Stores activities

Relationships

users 1:N contracts

contracts 1:N contract_versions

contracts 1:N obligations

contracts 1:N renewals

users 1:N notifications

users 1:N reports

users 1:N audit_logs

users 1:N activities
contracts 1:N activities

Working

The users table stores user information. Each user can manage multiple contracts. Every contract can have multiple versions, obligations, and renewals. Notifications keep users informed, reports provide summaries, audit logs record system actions, and activities track user interactions with contracts

ER Diagram

